EXIST 2026 Memes Dataset

Task 2 – Final Report

Sexism Identification, Source Intention, and Sexism Categorization in Memes

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Scope	EXIST 2026 Task 2 only: memes, subtasks 2.1, 2.2 and 2.3
Excluded scope	Videos and Task 3 out of scope by design
Dataset	3,984 training memes and 1,053 test memes
Modalities explored	OCR text, meme image, physiological signals and VLM-generated visual text
Submission limit	3 hard + 3 soft runs per subtask
Official metric	ICM for hard-hard evaluation and ICM-soft for soft-soft evaluation
Local proxy metric	Five-fold macro-F1 and macro-F1 on the official hard-gold subset

Generated from the experimental results and final submitted systems.

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1 Executive Summary

This work addresses the EXIST 2026 memes track for Task 2. The final systems produce hard and soft predictions for the three official subtasks:

Subtask	Problem type	Output space
2.1	Binary mono-label	`YES`, `NO`
2.2	Hierarchical mono-label	`NO`, `DIRECT`, `JUDGEMENTAL`
2.3	Hierarchical multilabel	`NO` plus five sexism categories

The final approach is VLM-enriched. Run 1 uses OCR plus a cached vision-language-model description and an explicit sexism analysis. Run 2 keeps only OCR plus the VLM visual description. Run 3 is a strong CPU-only backup selected from the best previous non-VLM runs. In all cases, the final classifier remains deliberately simple and reproducible: TF-IDF features with Logistic Regression, using One-vs-Rest Logistic Regression for the multilabel subtask.

Run	System	Input representation
1	VLM full	OCR text + VLM visual description + VLM sexism analysis
2	VLM description	OCR text + VLM visual description only
3	CPU backup	Task 2.1 uses OCR TF-IDF + physiological features; Tasks 2.2 and 2.3 use OCR TF-IDF text

The most defensible comparison is the five-fold cross-validation run over the same folds for the CPU and VLM variants:

Subtask	Best CPU CV	VLM desc CV	VLM full CV	Gain
2.1	0.6556	0.6809	**0.7179**	+0.0623
2.2	0.4531	0.4811	**0.5164**	+0.0633
2.3, 5-class	0.3645	0.4069	**0.4584**	+0.0939
2.3, 6-class	0.4215	0.4581	**0.5043**	+0.0828

Main finding
Physiological data provides small gains for the binary and source-intention subtasks, but it does not improve fine-grained multilabel categorization. The largest and most stable improvement comes from converting the meme image into semantically useful text with a VLM.

EXIST 2026 ranks systems with ICM and ICM-soft. Since the official test labels are not released, model selection inside the project uses macro-F1 on training-set validation as a transparent diagnostic proxy.

2 Official Evaluation Setting

The implementation follows the official EXIST 2026 lab guidelines. The task accepts hard outputs and soft probabilistic outputs. Hard outputs are evaluated against hard gold labels; soft outputs are evaluated against disagreement-aware soft gold labels. For all subtasks, the official ranking metric is ICM / ICM-soft, which is designed to handle hierarchy and learning with disagreement better than plain accuracy.

The number of submitted systems is also constrained: at most three hard and three soft runs are allowed per subtask. We use these slots as controlled ablations rather than as unrelated systems: a full VLM system, a description-only VLM system, and a CPU-only backup.

3 Dataset and Label Policy

The project uses only the meme split. The training set contains 3,984 memes and the test set contains 1,053 memes, balanced across Spanish and English. Test labels are not available and are never read by the pipeline.

The label handling mirrors the official learning-with-disagreement setup. Soft outputs are probability distributions derived from the model probabilities. Hard outputs are conventional labels for hard-hard evaluation. For Task 2.3, hard predictions are non-empty label lists, while soft predictions are per-label probabilities that do not need to sum to one because the task is multilabel.

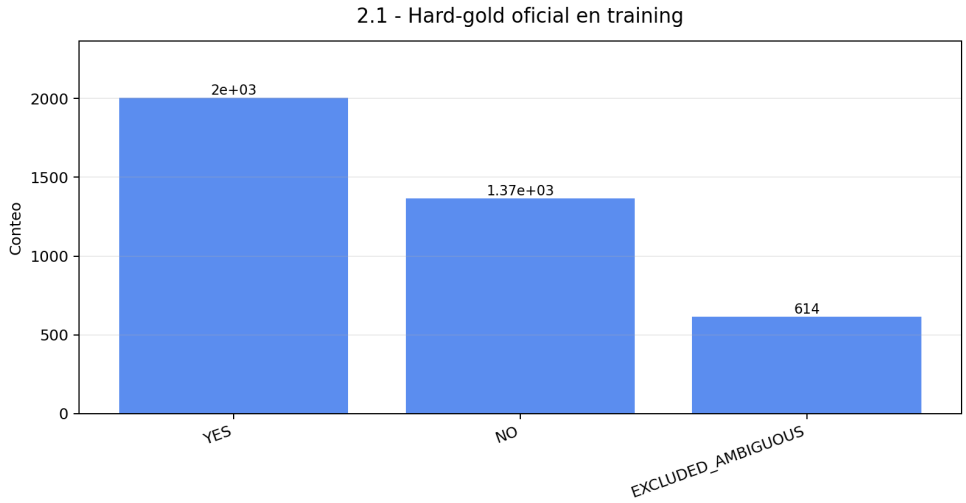


Figure 1: Task 2.1 training label distribution, including ambiguous items excluded from hard-hard evaluation.

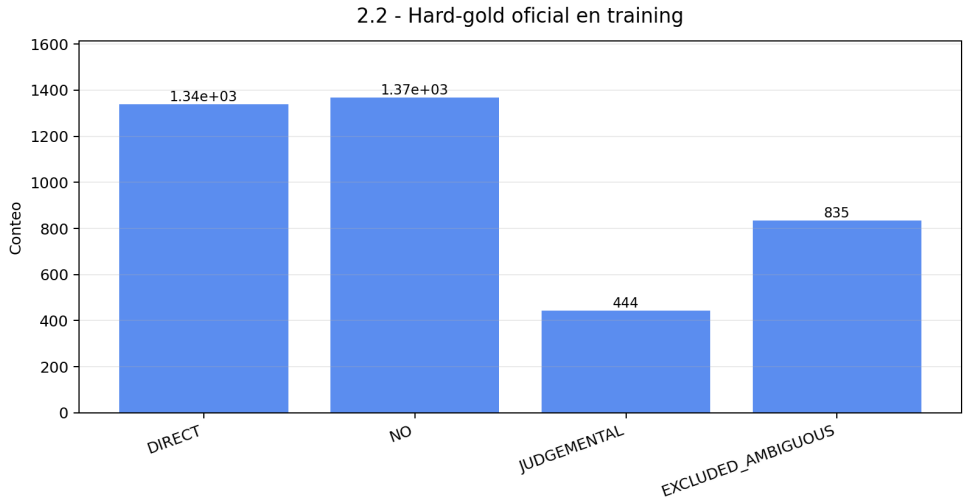


Figure 2: Task 2.2 training label distribution.

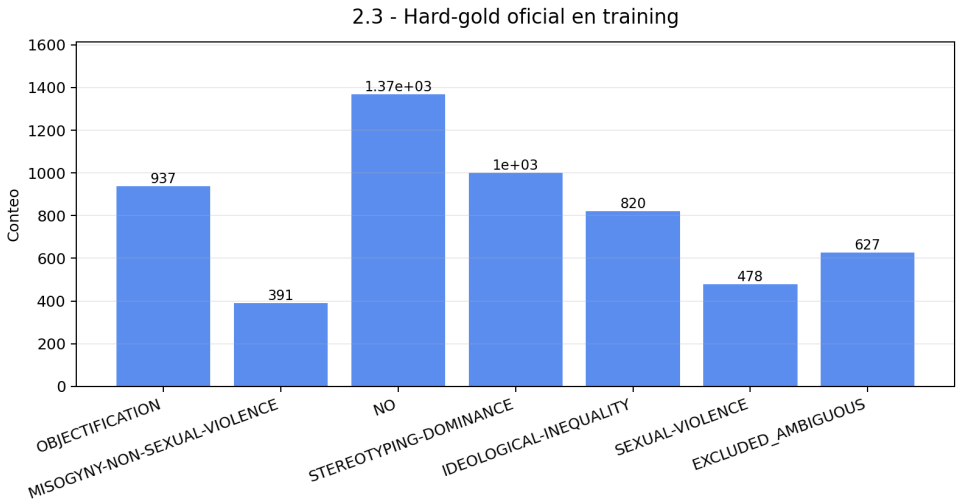


Figure 3: Task 2.3 multilabel training label distribution.

4 System Design

4.1 Text and VLM Enrichment

The baseline textual view uses the OCR text provided with each meme. Text is normalized with Unicode NFKC, lowercased, and cleaned for noisy social-media artifacts such as URLs and mentions. Stopwords are intentionally kept because gendered pronouns and short function words can carry signal in sexism detection.

The VLM track enriches OCR with generated textual records. Each record contains:

- a visual description of the meme image;
- an explicit sexism analysis;
- enough metadata to keep the generation process traceable.

This design does not make the VLM the final classifier. The VLM transforms visual information into text; the actual decision layer remains the same TF-IDF + Logistic Regression pipeline. This keeps the comparison clean: the main question is how much semantic visual text helps the classifier.

4.2 Feature Representation

All official runs use sparse TF-IDF features over text. The vectorizer concatenates:

- word n-grams, capturing lexical content and short expressions;
- character n-grams, improving robustness to OCR noise and multilingual morphology.

The CPU backup systems also use the earlier non-VLM feature experiments:

- **Physiological features**: Eye Tracking, Heart Rate and EEG summaries are median-imputed, clipped, standardized with train-only statistics and concatenated with their presence masks.
- **Low-level image features**: color, grayscale, histogram and edge descriptors were explored as an ablation, but they were weaker than both VLM and the strongest CPU text backups.

Signal group	Information captured	Expected contribution
Eye Tracking	Fixations, saccades, pupil and blink summaries	Attention and visual sampling patterns
Heart Rate	Arousal-related summary features	Emotional activation while viewing the meme
EEG	Band-power summaries over multiple channels	Cognitive load and affective response proxies

4.3 Classifiers

Tasks 2.1 and 2.2 use Logistic Regression with balanced class weights. Task 2.3 uses One-vs-Rest Logistic Regression to generate one probability per label. The hard multilabel output is obtained by thresholding probabilities; if no positive category crosses the threshold, the prediction is `NO`.

This design keeps the system inspectable: the semantic visual reasoning enters through VLM-enriched text fields, while the decision layer remains a simple linear classifier over reproducible TF-IDF features.

5 Run Mapping

Run ID	System	Role in the final submission
1	`vlm_v1`	Main system for all subtasks: OCR + VLM description + VLM sexism analysis
2	`vlmdesc_v1`	Ablation: OCR + VLM description only
3	CPU backup	Strongest previous CPU-only run for each subtask

Subtask	Run 3 source	Reason
2.1	TF-IDF + physiological features	Best CPU-only local system for binary sexism identification
2.2	TF-IDF text	Most stable CPU-only option for source-intention classification
2.3	TF-IDF text	Most stable CPU-only option for multilabel sexism categorization

The main runs are VLM-based because they dominate the CPU systems in cross-validation. The third run is kept as a transparent non-VLM comparison and as evidence of how far a CPU-only pipeline can go.

6 Validation Results

Model selection used reproducible local diagnostics because the official ranking is computed by the organizers with ICM/ICM-soft. The most comparable diagnostic is the five-fold cross-validation run over the same folds for CPU, VLM-description and VLM-full variants.

Subtask	Best CPU CV	VLM desc CV	VLM full CV	Gain vs CPU
2.1	0.6556	0.6809	**0.7179**	+0.0623
2.2	0.4531	0.4811	**0.5164**	+0.0633
2.3, 5-class	0.3645	0.4069	**0.4584**	+0.0939
2.3, 6-class	0.4215	0.4581	**0.5043**	+0.0828

The VLM full system is consistently the strongest across all subtasks. The description-only ablation already improves over the CPU baselines, and the explicit sexism-analysis field adds another clear gain, especially for the fine-grained multilabel categorization in Task 2.3.

6.1 Physiological Data: With vs Without

Because the 2026 dataset adds physiological signals, we explicitly tested whether those features improve the training-set cross-validation results. The comparison below uses the OCR-only TF-IDF baseline as the no-physiology system and OCR TF-IDF plus Eye Tracking, Heart Rate and EEG summaries as the physiology-enhanced system.

Task	OCR only	OCR + physiological	Difference	Interpretation
2.1	0.6430	0.6556	+0.0127	Small gain
2.2	0.4429	0.4531	+0.0102	Small gain
2.3, 5-class	0.3645	0.3573	-0.0073	Slight loss
2.3, 6-class	0.4215	0.4154	-0.0061	Slight loss

Physiological data helps the two mono-label tasks slightly. This is plausible: attention and arousal signals may correlate with whether a meme is perceived as sexist or with the broad source-intention decision. However, the same signal does not help Task 2.3. Fine-grained sexism categories require semantic evidence, and physiological responses do not directly indicate whether the meme expresses objectification, stereotyping, sexual violence or ideological inequality.

6.2 CPU Ablation Diagnostics

Before promoting the VLM track, three CPU-only systems were compared locally: OCR TF-IDF text, OCR TF-IDF plus physiological features, and OCR TF-IDF plus low-level image descriptors.

Subtask	Best CPU run	Macro-F1	Interpretation
2.1	TF-IDF + physiological	0.6940	Physiological signals slightly improved binary sexism identification.
2.2	TF-IDF text	0.4772	Pure text was more stable for source intention.
2.3	TF-IDF text	0.4578	Text was stronger than noisy auxiliary CPU features for multilabel categorization.

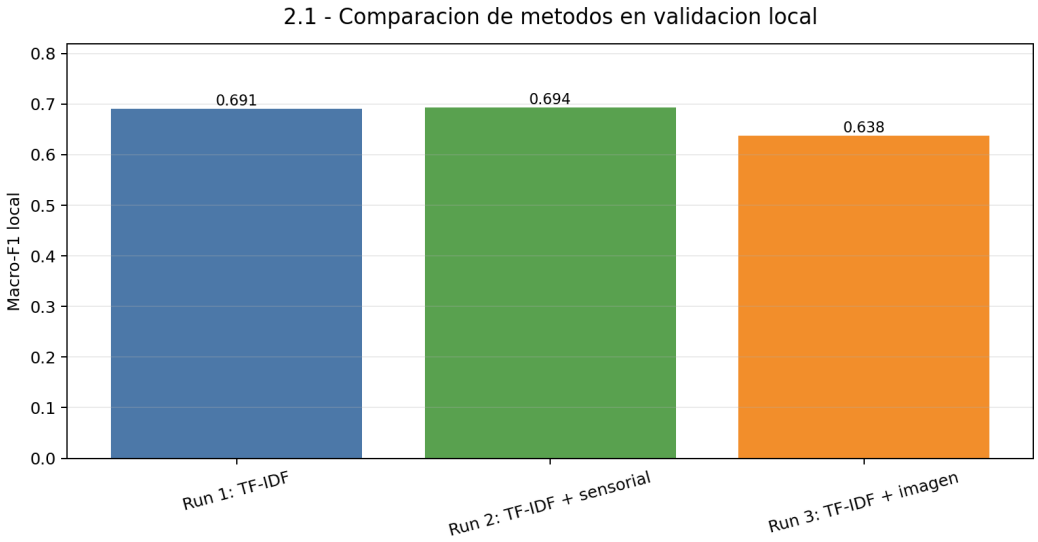


Figure 4: Task 2.1 CPU-only validation metrics used to select the backup run.

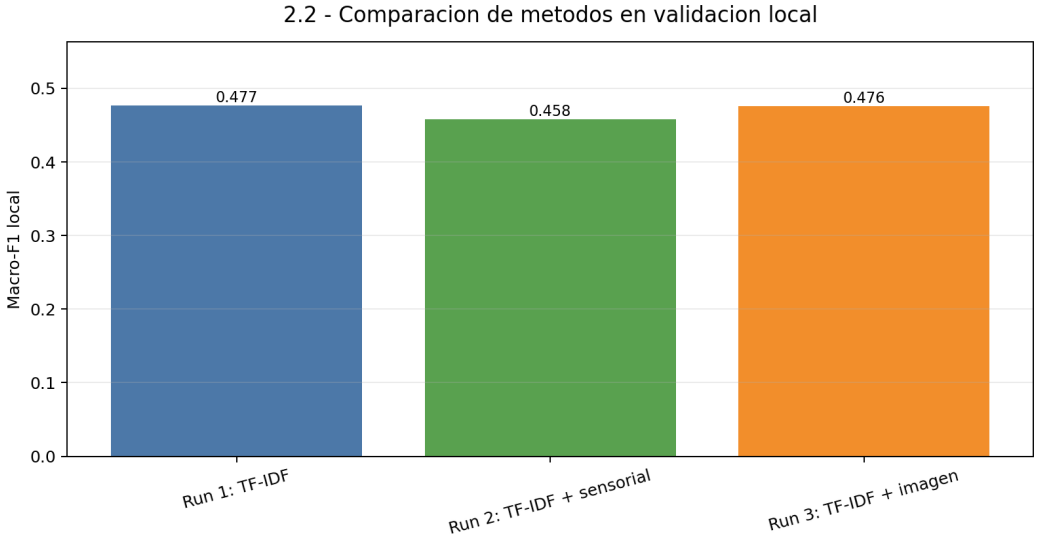


Figure 5: Task 2.2 CPU-only validation metrics used to select the backup run.

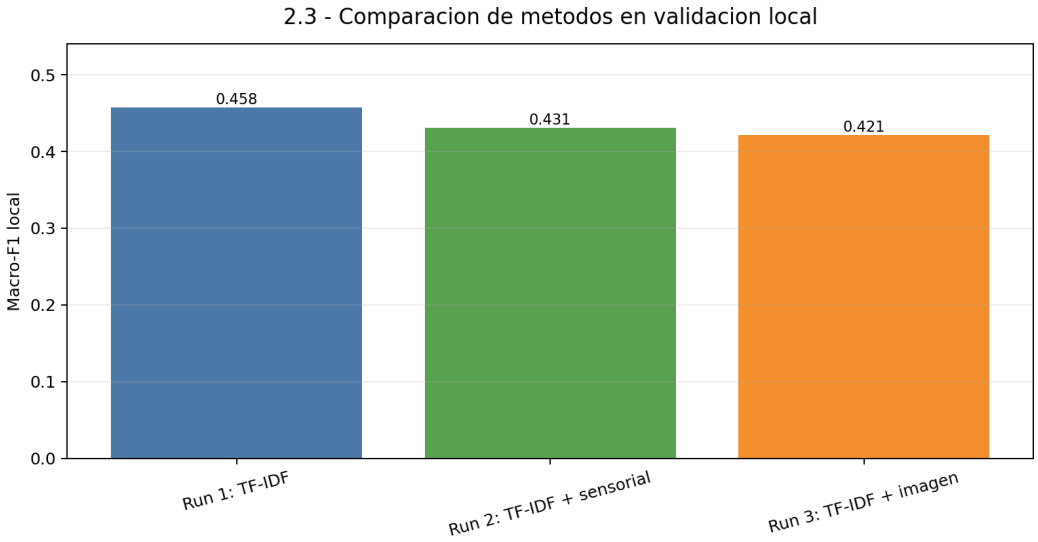


Figure 6: Task 2.3 CPU-only validation metrics used to select the backup run.

7 Discussion

The experiments support three main conclusions.

First, OCR text is a strong baseline. A simple TF-IDF representation with Logistic Regression is competitive because many memes contain explicit textual cues. Character n-grams are especially useful for noisy OCR and bilingual content.

Second, physiological data is informative but task-dependent. It improves Task 2.1 and Task 2.2 slightly, but it does not improve Task 2.3. This suggests that viewer response signals can help detect broad salience or perceived sexism, while fine-grained categories require semantic interpretation.

Third, VLM enrichment is the most useful visual component. Low-level image descriptors do not capture the meaning of the image, but a VLM description converts that image into text that can be exploited by the same classifier. Adding the explicit sexism analysis gives another consistent improvement, because the generated text contains concepts close to the target label space.

8 Limitations and Future Work

- The official ranking is ICM/ICM-soft. Local macro-F1 is only a transparent diagnostic proxy.
- The VLM-enriched representation depends on the quality of generated descriptions and analyses.
- Low-level image descriptors are reproducible but not semantic; they are useful as a baseline, not as a final visual representation.
- Transformer fine-tuning is plausible future work, but it should only be included if the final weights and experimental setup are reproducible.
- A future version should tune Task 2.3 thresholds directly against the official metric once local ICM/ICM-soft evaluation is available.

9 Conclusion

The final system is deliberately simple at the classifier level but rich at the representation level. Logistic Regression over TF-IDF remains competitive when the input text captures the right evidence. Physiological data contributes modestly, especially for binary sexism identification, but the strongest gains come from converting the meme image into semantically useful text. The final runs therefore prioritize VLM-enriched text, while preserving a CPU-only backup to document the effect of the new physiological modality.